ENVS 193DS Final

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GitHub Repo

Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # organizing files
library(rstatix) # statistical relationships
library(MuMIn) # modelling
library(DHARMa) # residual diagnostic simulation
library(ggeffects) # model predictions
library(ggplot2) # visualizations

# read in data nest data
nests <- read_csv(here("data", "nest_data_final.csv"))

# read in and make object for personal data
dinner <- read_csv(here("data", "Dinner_Data.csv"))
```

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, the researchers used a Generalized Linear Model (GLM) to reach their conclusion and compute that p-value, since the claim answers the central question of how distance to water edge predicts the probability of American Avocet microhabitat use. Additionally, we can make this conclusion because the variables are binary (use of microhabitat or no use) and continuous (distance from water edge) are non-parametric. Furthermore, since the response variable is binary we know the test was a logistic regression, a specific type of generalized linear regression. In part 2, they used a chi-square test because they are examining the relationship between two categorical variables, bird species and habitat type.

b. Suggestions for rewriting

Part 1.

We found that American Avocets do not use microhabitats in a random fashion, but rather with respect to distance to water edge. In this sense, distance to a water edge predictos the likelihood American Avocet use a microhabitat.

Part 2.

When looking at the distribution of American Avocet, Forster's Tern, and Black-necked Stilt, we found that each species was not spread randomly among habitats. Rather, each species presence differed based on habitat most likely due to assuming a certain niche in each habitat. This conclusion takes into consideration managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat as habitats.

c. Further Analysis

Another variable that could influence the probability of American Avocet microhabitat use is the predator abundance in a given habitat, which could be discrete numerical count data or ordinal categorical with pre determined levels from low abundance to high abundance of predators. This variable will influence American Avocet habitat use because this species might avoid areas with high concentrations of predators as they pose a risk for injury, mortality, and reduce reproductive success.

Problem 2. Data Analysis

a. Clean and Wrangle

```
# create new object, nest_clean
nests_clean <- nests |>
  # keep only height, bird species, and ant species
  select(height_cm, case_control, ant_species) |>
  # filter to include on C. mimosae, C. sjostedti, and C. nigriceps
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create new column for scientific species names
  mutate(scientific_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti", #C. sjostedti
    "RRB" ~ "Crematogaster mimosae", # C. mimosae
    "BBR" ~ "Crematogaster nigriceps" # C. nigriceps
  )) |>
  # order the species
  mutate(scientific_name =
           as_factor(scientific_name), # factor by name
         # level by scientific name
         scientific_name = fct_relevel(scientific_name,
                                       "Crematogaster sjostedti",
                                       "Crematogaster mimosae",
                                       "Crematogaster nigriceps"))

# display new structure of the data
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```
# display 10 random rows from data
slice_sample(nests_clean, n =10)
```

```
# A tibble: 10 x 4
   height_cm case_control ant_species scientific_name
```

```
       <dbl>        <dbl> <chr>       <fct>
 1      238.            0 RRB         Crematogaster mimosae
 2      212.            1 RRB         Crematogaster mimosae
 3      410.            1 RRB         Crematogaster mimosae
 4      191             1 RRB         Crematogaster mimosae
 5      513             0 RRB         Crematogaster mimosae
 6      228             0 RRB         Crematogaster mimosae
 7      114.            0 RRB         Crematogaster mimosae
 8      270.            0 RRB         Crematogaster mimosae
 9      152             0 RRB         Crematogaster mimosae
10      150.            0 RRB         Crematogaster mimosae
```

b. Response Variable

The 1s and 0s in the case_control column indicates whether or not a tree contained a nest. If a data point has a 1 this indicates the tree contains a nest, whereas a 0 indicates the tree did not contain a nest and was "available" for comparison.

c. Purpose of Study

Tree Height

Tree height is a continuous variable measured in cm. This variable can influence the probability of nest occurrence as tree height increases the probability of nest occurrence also increases because they might provide more shelter from predators or procide necessary vantage points to identify food sources. This is supported by the fact that tree architecture plays an important role in nest site selection (Discussion, Paragraph 2) and that after running a logistic regression model it was found that each 1 m increase in height increased the odds of a bird nesting in a tree by 20% ("Results", Paragraph 2)

Acacia Ant Species

Acacia Ant Species is a categorical variable, broken into three species C. mimosae, C. sjostedti, and C. nigriceps. The articles discusses how the presence of aggressive ants could cause decreasing in the number of nests due to the fact that these are nest predators and might inhibit their ability to find food ("Introduction", Paragraph 3); however, they also discuss how the presence of aggressive ants might increase the probability of nesting occurences because they ward off predators making these trees safer for bird to live and raise offspring ("Introduction", Paragraph 3)

d. Table of Models

Model number	Ant Species	Tree Height	Model description
0	X	X	All predictors (No Interaction term, +)
1	X	X	All predictors (Interaction term, *)

e. Run the Models

```
# model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ scientific_name + height_cm,
                   # no interaction model variables
                   data = nests_clean, # use clean dataset
                   family = binomial) # set to binomial

# model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ scientific_name * height_cm,
                   # interaction model variables
                   data = nests_clean, # use clean data set
                   family = binomial) # set to binomial
```

f. Select the Best Model

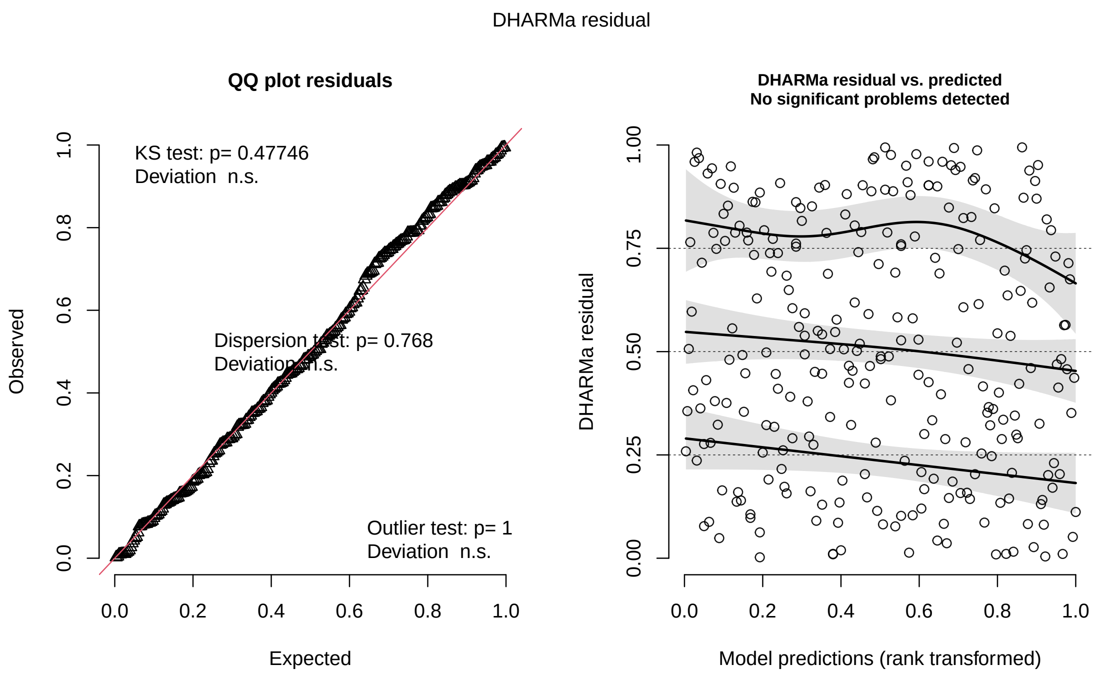
```
# calculate AIC for two models
AICc(
  nest_model0, # model 0 no interaction
  nest_model1 # model 1 interaction
) |>
  # arranging outputs in order from lowest to highest
  arrange(AICc)
```

```
            df     AICc
nest_model1  6 238.1236
nest_model0  4 258.8940
```

The best model for predicting the probability of bird nest presence as determined by Akaike's Information Criterion (AIC) is the model of interaction between tree height (cm) and ant species, demonstrated by the lower AIC value.

g. Check the Diagnostics

```
# check diagnostic with DHARMa
plot(
  simulateResiduals(nest_model1) # set to model 1
)
```



The QQ plot indicates that there is a uniform distribution of the data, since the points follow the reference line. On the right is the residuals vs. predicted plots. Since the residuals have no distinct pattern, they are randomly scattered and have a random distribution of residuals. The assumption to use a generalized linear model is met

h. Visualize the Model Predictions

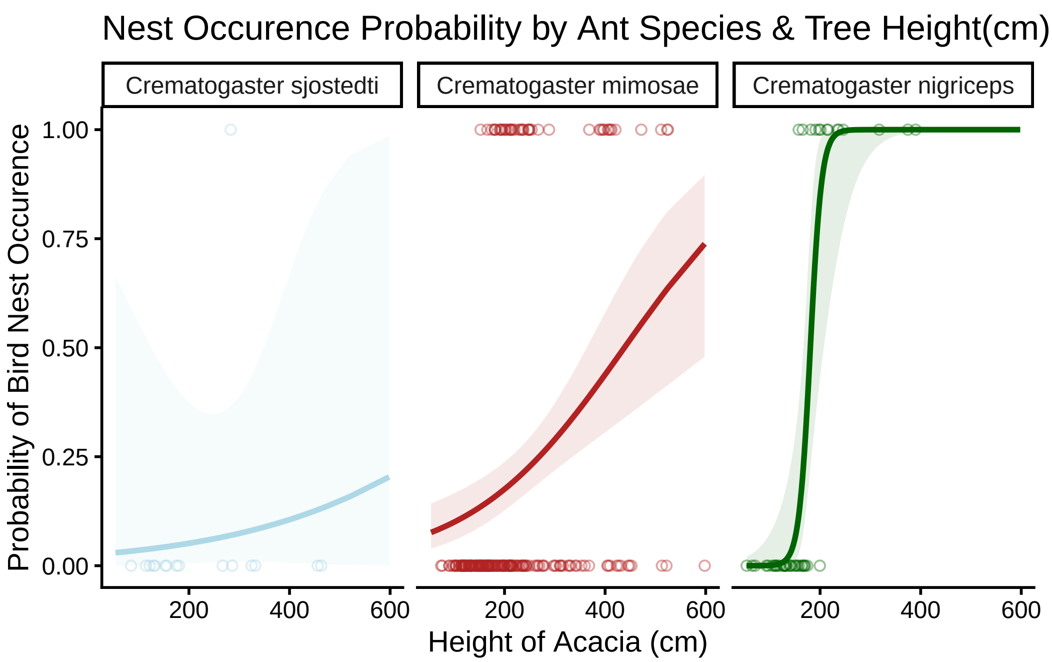
```
# create preds object
preds <- ggpredict(
  nest_model1, # model object 1
  terms = c("height_cm[all]", "scientific_name")
  # predictors height and scientific name
```

```
  )

# add scientific_name column for wrap in visualization
preds$scientific_name <- preds$group
```

```
# base layer: ggplot
ggplot(data = nests_clean, # set data to nest_clean
       mapping = aes(x = height_cm, # x-axis height
                     y = case_control, # y-axis nest occurence
                     color = as.factor(scientific_name))) +
  # color based on ant name
  # relabelling x- and y-axis
  labs(x = "Height of Acacia (cm)", # x-axis height
       y = "Probability of Bird Nest Occurence", # y-axis nest probability
       color = "scientific_name", # color by ant species
       title =
         "Nest Occurence Probability by Ant Species & Tree Height(cm)")+
  # add tile
  # manually change ant colors
  scale_color_manual(values = c("Crematogaster sjostedti" = "lightblue",
                               "Crematogaster mimosae" = "firebrick",
                               "Crematogaster nigriceps" = "darkgreen")) +
  # fill color based on species
  scale_fill_manual(values = c("Crematogaster sjostedti" = "lightblue",
                               "Crematogaster mimosae" = "firebrick",
                               "Crematogaster nigriceps" = "darkgreen")) +
  # change theme from default
  theme_classic() +
  # remove gridlines and background
  theme(panel.background = element_rect()) +
  # remove legend
  theme(legend.position = "none") +
  # first layer: geom points, showing underlying data
  geom_point(mapping = aes(x = height_cm, # x-axis: aciacia height
                           y = case_control, # y-axis: nest occurence
                           color = scientific_name), # set to selected colors
             shape = 21, # change shape
             alpha = 0.4,# make slightly transparent
             inherit.aes = FALSE) + # ignore global aes for colors
  # second layer: geom line, showing predictions
  geom_line(data = preds, # data from model 1 prediction
            mapping = aes(x = x, # x-axis height of acacia
```

```
                          y = predicted, # y axis: prediction probability
                          color = group), # color by ant
            linewidth = 1) +
  # third layer: ribbon (confidence interval)
  geom_ribbon(data = preds, # data from model 1 prediction
              mapping = aes(x = x, # x-axis height of acacia
                            ymin = conf.low, # set low value
                            ymax = conf.high, # set high value
                            fill = group), # fill by ant species
              inherit.aes = FALSE, # ignore global aes
              alpha = 0.1) + # make transparent
  # make new panel for each ant species
  facet_wrap(~ scientific_name)
```



i. Write a Caption for your Figure

Figure 1. Probability of bird nest occurrence differs depending on the ant species present and acacia tree height Each figure separates the data, the prediction of nest occurrence given tree height (cm), and the 95% confidence interval based on ant species. The blue figure depicts data associated with the Crematogaster sjostedti, the red figure depicts data associated with Crematogaster mimosae, and the green figure represents data

associated with Crematofaster nigriceps. Circles represent individual observations of nest occurrences in each scenario, the lines represent model predictions of the probability of nest occurrences given tree height (cm) and ant species, and the ribbon represents the 95% confidence interval. Data originally from, Lujan, E., et al. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna. Zenodo. https://doi.org/10.5281/zenodo.8373322

j. Calculate Model Predictions

```
# running ggpredict for model prediction at height of 600cm
ggpredict(nest_model1, # use model 1 w/ interaction
          terms = c("height_cm [600]", # 1st term = height 600 cm
                   "scientific_name")) # 2nd term = ant species
```

```
# Predicted probabilities of case_control

scientific_name: Crematogaster sjostedti

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

scientific_name: Crematogaster mimosae

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

scientific_name: Crematogaster nigriceps

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your Results

According to the results from part j., at tree height of 600 cm, the probability of bird nest occurrence with Crematogaster sjostedti present is 0.2 (95% CI: [0.00, 0.99]), with Crematogaster mimosae present is 0.74 (95% CI:[0.48, 0.90]), and with Crematogaster nigriceps present is 1.00 (95% CI: [1.00, 1.00]). According to both figure 1. and the prediction probabilities from the output of part j., there is a relationship between birds' preference to nest in taller trees inhabited by Crematogaster mimosae and Crematogaster nigriceps as nesting probability increases steeply with tree height for both species. Figure 1. demonstrates how the prediction probability curve increases steeply as tree height increases for C. mimosae and C. nigriceps which is also reflected in the predicted probabilities being large values of 0.74 and 1.00 respectively at 600 cm; however, when observing the C. sjostedti the figure remains relatively flat across all heights and only reaches a probability of 0.2 at 600 cm in part j. Biologically, this preference for trees inhabited by aggressive ant species (Crematogaster mimosae and Crematogaster nigriceps) occurs because the ants do not directly interfere with adults and offspring and actually reduce the risk of bird predation, whereas the Crematogaster sjostedti only provides minimal protection. There is a greater preference towards Crematogaster nifriceps because this species prunes, shortening shoots and creates defensive canopies that provide further protection for birds from predators like snakes, mesocarnivores, and raptors.

Problem 3. Affective and Exploratory Visualizations

a. Comparing Visualizations

How are the visualizations different from each other in the way you have represented your data?

My affective visualization was extremely different from my initial exploratory visualization in Homework 2 as the data represented and the formats were drastically altered. I originally chose to create a histogram showing count data of where dinner was cooked over the different days of the week and the background scatterplot for a generalized linear model that demonstrated the probability of cooking at home on the y axis and hours worked on the x axis; however, my affective visualization created individual figures that depicted the hours worked and whether I cooked at home or not depending on that specific day. While my original visualization had no connection between time, whether I cooked at home or ate out, and hours I worked, the affective visualization was a calendar and incorporate data on the day of the week, with images that depicted hours worked and whether I cooked or ate out.

What similarities do you see between all your visualizations?

At its core, these visualizations depict individual data points that give insight of how many hours I worked in one day and whether or not I cooked dinner at home that day. Whether its a open circle on the scatter plot background of a GLM model or the number of cracks on one plate holding dinner or a takeout bag, these styles reflect an overall trend that connects my hours working with whether I cooked at home or not.

What patterns (e.g. differences in means/counts/propotions/medians, trends through time, relationships between variables) do you see in each visualization? Are these different between visualizations? If so, why? If not, why not?

Both visualizations show the same pattern, as hours worked increases the proportion of days I ordered food out increases (and cooking at home decreases). In the explanatory visualization, this is shown by most data points on the probability line of 1.0 being clustered around low hours worked and high counts on the 0.0 line clustering at high hours worked, whereas my affective visualization portrayed a similar relationship through increased cracks on plates (hours worked) predominantly on plates with takeout bags. The patterns seem similar to one another because both draw from the same data set, the only difference is the style in which I communicate these specific data points.

What kinds of feedback did you get during week 9 workshop or from the instructors? How did you implement or try those suggestions? If you tried and kept those suggestions, explain how and why; if not, explain why not?

During workshop 9 my main feedback was to include more of the explanatory variables that I kept tract of in my affective visualization, rather than just using hours worked and whether I cooked dinner or not. I implemented these suggestions by adding more novel characteristics to my calendar like color of the plate to represent whether I grocery shopped and number of prongs on a fork to indicate how many assignments I had due that day. I ended up keeping these suggestions by recreating my visualization and drawing these factors in with colored pencil because I felt as though they added more depth to my research question and demonstrated how other variables might have an affect on my dinner habits, not just hours I worked.

b. Designing an Analysis

My primary research question was "How does daily work hours relate to my decision to cook dinner at home or eat out?" My predictor variable was hours worked which is a continuous variable and my response variable was whether or not I cooked at home, an example of categorical, binary data.

Given my central question and my variables of interest, an analysis that I could apply to my research question is a generalized linear model with a binomial error distribution. This analysis makes sense because my central question is examining how one variable (hours worked) affects the likelihood of another variable (whether I cook at home or not). Additionally, since the response variable is binary a logistic regression fits best.

c. Check any Assumptions and Run your Analysis

```
# store dinner data as dinner_clean
dinner_clean <- dinner |>
  # clean names, making lowercase and underscore
  clean_names() |>
  # name treatments
  mutate(cooked_at_home_y_n = as.numeric(case_when(
    # mutate yes to "1"
    cooked_at_home_y_n == "Yes" ~ "1",
    # mutate no to "0"
    cooked_at_home_y_n == "No" ~ "0")))
```

As a small side note, I would like to acknowledge that when running GLMs it is common practice to make model tables to demonstrate the various interactions between variables and run an AIC to find which model fits the data best; however, after speaking with Professor Bui I determined since I am only examining the affect between two variables these were unnecessary.

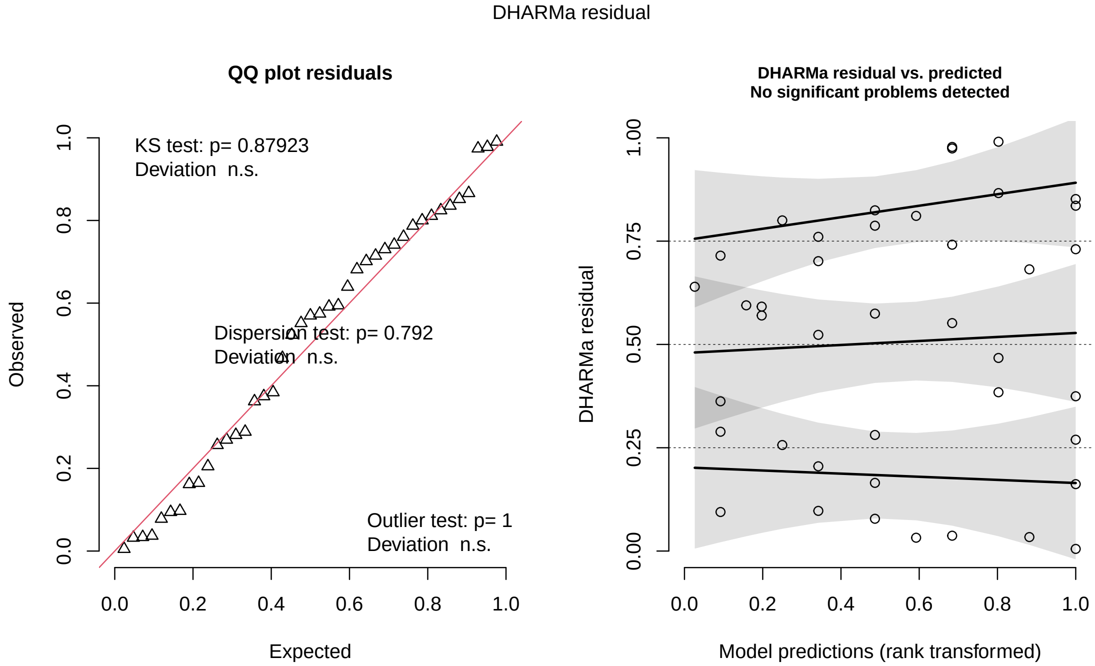
Additionally, since my data points were independent observations my data meets that assumption for ther analysis. Furthermore, the response variable is binomial meaning it is distributed following an exponential family distribution, meeting another assumption

Write in Model

```
# create model for where I cooked and time worked
dinner_mod <- glm(cooked_at_home_y_n ~ time_worked_hours,
                  # set variables of interest
                  data = dinner_clean,# set data to dinner_clean
                  family = "binomial") # binomial family distribution
```

Check the Diagnostics with DHARMa

```
# run diagnostic check using DHARMa package
plot(
  simulateResiduals(dinner_mod) # use dinner model
)
```



The DHARMa diagnostic, demonstrates through the QQ plot that the residuals follow a uniform distribution since the points follow the red guideline. Additionally the dispersion test and outlier test, tell us no residuals experience a significant amount of dispersion or act as outlines impacting results. Additionally the right figure tells us that no significant problems were detected between the residuals and predicted. The assumptions to use a generalized linear model are met.

Model Coefficients and Calculating Probabilities

```
# create a summary table of models coefficients
summary(dinner_mod)
```

```
Call:
glm(formula = cooked_at_home_y_n ~ time_worked_hours, family = "binomial",
    data = dinner_clean)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)         1.8462     0.7069   2.612  0.00901 **
```

```
time_worked_hours  -0.3039     0.1479  -2.055  0.03984 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 52.644  on 40  degrees of freedom
Residual deviance: 47.808  on 39  degrees of freedom
AIC: 51.808

Number of Fisher Scoring iterations: 4
```

```
# calculate probability of cooking at home when I work 0 hours
plogis(1.8462)
```

```
[1] 0.8636803
```

```
# calculate odds ratio
exp(-0.3039)
```

```
[1] 0.7379347
```

```
# confidence intervals of odds ratio
exp(confint(dinner_mod))
```

```
                      2.5 %     97.5 %
(Intercept)       1.7926414 30.2930849
time_worked_hours 0.5356862  0.9687707
```

```
# calculate the probability of cooking at home after working 8 hours
ggpredict(dinner_mod, # set model
          terms = "time_worked_hours [8]") # set hours worked to 8
```

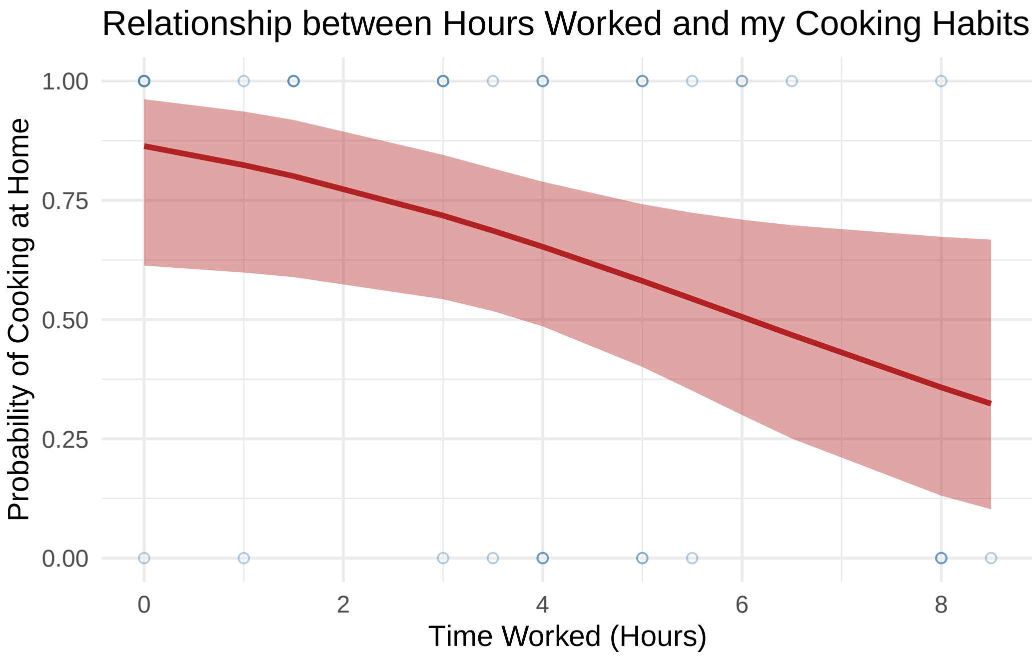
```
# Predicted probabilities of cooked_at_home_y_n

time_worked_hours | Predicted |     95% CI
------------------------------------------
                8 |      0.36 | 0.13, 0.67
```

d. Create a Visualization

```
# create pred_dinner object
preds_dinner <- ggpredict(
  dinner_mod, # use dinner model
  terms = c("time_worked_hours [all]")
) # predictor is time worked

# base layer: ggplot
ggplot(data = dinner_clean, # data from dinner_clean
       mapping = aes(x = time_worked_hours, # x-axis time worked
                     y = cooked_at_home_y_n)) + # y-axis whether I cooked
  # relabelling x- and y- axes
  labs(x = "Time Worked (Hours)", # x-axis time worked
       y = "Probability of Cooking at Home", # y-axis probability of cooking
       title = "Relationship between Hours Worked and my Cooking Habits") +
  # create title
  # first layer: geom points showing data
  geom_point(color = "steelblue", # change color
             alpha = 0.4, # change opacity
             shape = 21) + # make open circles
  # second layer: geom line showing probability
  geom_line(data = preds_dinner, # set data to prediction
            mapping = aes(x = x, # x variable
                          y = predicted), # y variable is predicted
            color = "firebrick", # change color
            linewidth = 1, # alter line width
            inherit.aes = FALSE) + # don't use ggplot aes
  # third layer: ribbon (95% CI Prediction)
  geom_ribbon(data = preds_dinner, # data from prediction
              mapping = aes(x = x, # set x axis data
                            ymin = conf.low,
                            # lower confidence interval for ribbon
                            ymax = conf.high),
              # upper confidence interval for ribbon
              fill = "firebrick", # color of CI
              alpha = 0.4, # opacity of CI
              inherit.aes = FALSE) +# don't use ggplot aes
  theme_minimal() # change theme
```



e. Write a Caption

Figure 2. Probability of cooking dinner differs based on hours worked. The figure depicts the individual data points, predicted probability based on hours worked, and the confidence interval associated with the probability that I cook at home depending on how much I work. 1 indicates I cook at home and 0 indicates I eat out. Small blue open circles represent individual observations of whether I cooked at home or not in relation to the amount I worked that day. The bold red line represents the model prediction of cooking at home given time worked(hours). Finally the red ribbon represents the 95% confidence interval. Data collected by Stephen Bond from 2026-04-20 to 2026-05-30.

f. Write About Your Results

The probability of cooking at home when time worked is 0 hours is 0.86. With every 1 hour increase in time worked, the odds of me cooking at home decrease by a factor of 0.74(95% CI:[0.536, 0.969], p = 0.04, $\alpha = 0.05$). When working 8 hours, the probability of me cooking at home is 0.36(95% CI:[0.13, 0.67]). The model coefficient of 0.74 from the odds ratio summary along with the predicted probability curve from Figure 2. decreasing from 0.86 at 0 hours to 0.36 at 8 hours, indicate that I tend to cook dinner at home more often when I work fewer hours.

g. Sharing Your Affective Visualization

I was present in workshop!